Math 118: Mathematical Methods of Data Theory

Lecture 5: The Singular Value Decomposition.

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Singular Values

- **Recall:** λ and $\boldsymbol{v}$ are an eigen- value/ vector pair for $A \in \mathbb{R}^{n \times n}$ if $A\boldsymbol{v} = \lambda \boldsymbol{v}$
- **Question:** Can we define something similar for non-square matrix $A \in \mathbb{R}^{m \times n}$?
- We say σ is a **singular value** of A if exist $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{u} \in \mathbb{R}^m$ such that $A\boldsymbol{v} = \sigma \boldsymbol{u}$ and $A^\top \boldsymbol{u} = \sigma \boldsymbol{v}$
- $\boldsymbol{u}$ is called a (left) **singular vector** and $\boldsymbol{v}$ is called a (right) **singular vector**.

Theorem

If σ, $\boldsymbol{u}$ and $\boldsymbol{v}$ are a singular value/ vector triple for $A \in \mathbb{R}^{m \times n}$ then σ^2, $\boldsymbol{v}$ eigen-value/vector pair for $A^\top A$ and σ^2, $\boldsymbol{u}$ an eigen- value/vector pair for $AA^\top$

Proof.

$$A\boldsymbol{v} = \sigma \boldsymbol{u} \Rightarrow A^\top A\boldsymbol{v} = \sigma A^\top \boldsymbol{u} = \sigma\left(\sigma \boldsymbol{v}\right) = \sigma^2 \boldsymbol{v}$$
$$A^\top \boldsymbol{u} = \sigma \boldsymbol{v} \Rightarrow AA^\top \boldsymbol{u} = \sigma A\boldsymbol{v} = \sigma\left(\sigma \boldsymbol{u}\right) = \sigma^2 \boldsymbol{u}$$

□

The Singular Value Decomposition

Recall: If $A \in \mathbb{R}^{n \times n}$ normal then $A = U\Lambda U^\top$.

- **Now:** $A \in \mathbb{R}^{m \times n}$ with $m \geq n$.
- **Theorem:**
  1. A has n singular values $\sigma_1, \ldots, \sigma_n$. Let $\Sigma = \text{diag}[\sigma_1, \ldots, \sigma_n] \in \mathbb{R}^{n \times n}$.
  2. $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n \in \mathbb{R}^m$ orthonormal. Let $U_1 = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n] \in \mathbb{R}^{m \times n}$
  3. $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{R}^n$ orthonormal. Let $V = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n] \in \mathbb{R}^{n \times n}$
- **Observe:** $AV = [A\boldsymbol{v}_1, \ldots, A\boldsymbol{v}_n] = [\sigma_1\boldsymbol{u}_1, \ldots, \sigma_n\boldsymbol{u}_n] = U_1\Sigma$
- **So:** $A = AVV^\top = U_1\Sigma V^\top$.

Theorem

Any $A \in \mathbb{R}^{m \times n}$ *with* $m \geq n$ *has:*

1. **Thin SVD:** $A = U_1\Sigma V^\top$.
2. **SVD:** $A = U \begin{bmatrix} \Sigma \\ 0 \end{bmatrix} V^\top$ *with* $U = [U_1, U_2] \in \mathbb{R}^{m \times m}$ *orthogonal.*

Finding the SVD

- Always assume singular values ordered: $\sigma_1 \geq \sigma_2 \geq \ldots \sigma_n \geq 0$.
- Exist $O(mn^2)$ algorithms[1] for finding U, Σ, V. (MATLAB: `svd`)
- Exist $\approx O(mn \log(m/\epsilon)$ algorithms[2] to find ϵ-good approximation to $\sigma_1, \boldsymbol{u}_1, \boldsymbol{v}_1$. (MATLAB: `svds`)
- Exist randomized algorithms [3] that approximately find $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_k$ and $\sigma_1, \ldots, \sigma_k$ in (expected) $O(mn \log(k) + (m+n)k^2)$ time.

[1] Use QR decomposition

[2] Keywords: Lanczos or Power methods

[3] See "Ten Lectures and Forty-Two Open Problems in Data Science" by Afonso Bandeira

Application 1: 2 norm of a matrix

- $\|A\|_2 = \sigma_1$ (largest singular vector).

Proof.

1. *Unitary invariance:* $\|A\|_2 = \|U\Sigma V^\top\|_2 = \|\Sigma\|_2$
2. *Norm of a diagonal matrix = largest entry:*

$$\|\Sigma\|_2 = \max_{\|\boldsymbol{x}\|_2=1} \|\Sigma\boldsymbol{x}\|_2 = \max_{\|\boldsymbol{x}\|_2=1} \sqrt{\sum_{i=1}^{m} \sigma_i^2 x_i^2}$$

$$\leq \max_{\|\boldsymbol{x}\|_2=1} \sqrt{\sum_{i=1}^{m} \sigma_1^2 x_i^2} = \sigma_1 \max_{\|\boldsymbol{x}\|_2=1} \sqrt{\sum_{i=1}^{m} x_i^2} = \sigma_1 \max_{\|\boldsymbol{x}\|_2=1} \|\boldsymbol{x}\|_2 = \sigma_1$$

and:

$$\|\Sigma\|_2 = \max_{\|\boldsymbol{x}\|_2=1} \|\Sigma\boldsymbol{x}\|_2 \geq \left\| \Sigma \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\|_2 = \left\| \begin{bmatrix} \sigma_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\|_2 = \sigma_1$$

□

Application 2: Rank and Low Rank Approximation

Rank of $A \in \mathbb{R}^{m \times n}$ is **dimension of range**.

- **Outer Product Form:**

$$A = U_1 \Sigma V^\top = U_1 \Sigma_1 \begin{bmatrix} \boldsymbol{v}_1^\top \\ \vdots \\ \boldsymbol{v}_n^\top \end{bmatrix} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n] \begin{bmatrix} \sigma_1 \boldsymbol{v}_1^\top \\ \vdots \\ \sigma_n \boldsymbol{v}_n^\top \end{bmatrix} = \sum_{i=1}^{n} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^\top$$

- $\boldsymbol{u}_i \boldsymbol{v}_i^\top \in \mathbb{R}^{m \times n}$ is a **rank-one matrix**. Why? Any $\boldsymbol{x} \in \mathbb{R}^n$:

$$\left(\boldsymbol{u}_i \boldsymbol{v}_i^\top\right) \boldsymbol{x} = \boldsymbol{u}_i \left(\boldsymbol{v}_i^\top \boldsymbol{x}\right) = \underbrace{\left(\boldsymbol{v}_i^\top \boldsymbol{x}\right)}_{\text{scalar}} \boldsymbol{u}_i \in \mathbb{R}^m$$

- **Question:** How to tell if A is rank r?
- **Answer:** $\sigma_r > 0$ but $\sigma_{r+1} = \ldots = \sigma_n = 0$.
- **Question:** What is best rank r approximation to A?
- **Answer:** $A_r := \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^\top = U_1 \Sigma_r V^\top$

Low Rank Approximation

Theorem

Consider the **Low Rank Approximation Problem** $Z^* = \underset{\text{rank}(Z)=r}{\arg\min} \|A - Z\|_2$.

Then $Z^* = A_r = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^\top$

Remark: Also true for Frobenius norm $\|A\|_F$.

Application 3: PCA for Movie Reviews

- Consider 6 anonymous users $P_1, \ldots, P_6$ and 4 movies $M_1, \ldots, M_4$. Construct matrix $X \in \mathbb{R}^{6\times 4}$ where X_{ij} is P_i's review of M_j.
- **Mean:** $\boldsymbol{\mu} \in \mathbb{R}^6$ where $\mu_i = \frac{1}{6}\sum_{j=1}^{6} X_{ij}$.
- **Center the Data** $\tilde{X} = X - \boldsymbol{\mu}\mathbf{1}^\top$.
- **Take the SVD:** $\tilde{X} = U\Sigma V^\top$. Right singular vectors $\boldsymbol{v}_i \in \mathbb{R}^4$ called principal components.
- **Interpretation:**
  1. $\boldsymbol{v}_i$ represents category of user, *eg* $\boldsymbol{v}_1$ represents "horror enthusiasts", $\boldsymbol{v}_2$ represents "children".
  2. v_{ij} represents "loading" or correlation of M_j with category i .
  3. $\boldsymbol{u}_j$ represents category of movie, $\boldsymbol{u}_1$ represents "horror" and $\boldsymbol{u}_2$ represents "children's movie".
  4. u_{ij} represents correlation of P_i with category j.
  5. $\frac{\sigma_1}{\sum_{i=1}^{4}\sigma_i}$ represents proportion of variance explained by $\boldsymbol{v}_1$, *eg* if all movies horror, then $\frac{\sigma_1}{\sum_{i=1}^{4}\sigma_i} \approx 1$.
- Usually, only a few principal components explain **most of the variance** in the data.

Application 4: Perturbation Analysis for Least Squares

- **Recall:** for square $A \in \mathbb{R}^{n \times n}$ condition number is $\kappa(A) = \|A\|_2 \|A^{-1}\|_2$
- For **non-square** $A \in \mathbb{R}^{m \times n}$ condition number is $\kappa(A) = \frac{\sigma_1}{\sigma_r}$ where $r = \text{rank}(A)$.

Theorem

1. *Suppose* $\boldsymbol{x}^* = \arg\min \|A\boldsymbol{x} - \boldsymbol{b}\|_2$ *and* $\hat{\boldsymbol{x}} = \arg\min \|(A + \delta A)\boldsymbol{x} - (\boldsymbol{b} + \delta \boldsymbol{b})\|_2$
2. *Let* $\epsilon_A = \dfrac{\|\delta A\|_2}{\|A\|_2}$ *and* $\boldsymbol{r} = \boldsymbol{b} - A\boldsymbol{x}^*$.
3. *Then:* $\|\boldsymbol{x}^* - \hat{\boldsymbol{x}}\|_2 \leq \dfrac{\kappa(A)}{1 - \epsilon_A \kappa(A)} \left(\epsilon_A \|\boldsymbol{x}^*\|_2 + \dfrac{\|\delta \boldsymbol{b}\|_2}{\|A\|_2} + \epsilon_A \kappa(A) \dfrac{\|\boldsymbol{r}\|_2}{\|A\|_2} \right)$